MATH 1352-009 Exam III November 22, 2005

Answer the problems on separate paper. You do <u>not</u> need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **<u>Show all relevant supporting steps!</u>**

Bald solutions to problems – answers without accompanying, supporting work – will receive ***<u>no</u>*** credit.

For each problem choose **1** (one, uno, eins, un) of the two options.

1. (10 pts) Choose one. Find $\frac{dy}{dx}$. Simplify where possible.

   a. $y = \operatorname{sech}(\frac{1+x^2}{1-x^2})$ b. $y = \tanh^{-1}(1/x)$

2. (10 pts) Choose one. Compute the integral.

   a. $\int \frac{\tanh(\sqrt{x})}{\sqrt{x}}\, dx$ b. $\int \frac{16x}{\sqrt{1+4x^4}}\, dx$

3. (10 pts) Choose one. Compute the limit of the sequence, where it exists. Show all supporting work.

   a. $\left\{\frac{4-7n}{8+2n^2}\right\}$ b. $\left\{\frac{\ln n^2}{n}\right\}$

4. (16 pts) Choose one. Determine whether the series converges or diverges. Show all supporting work.

   a. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2n+5}}$ b. $\sum_{n=1}^{\infty} \frac{(n+4)^2}{n^{\pi}+1}$

5. (16 pts) Choose one. Determine whether the series converges or diverges. Show all supporting work.

   a. $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$ b. $\sum_{n=1}^{\infty} \frac{n^2}{3n^2+2}$

6. (16 pts) Choose one. Determine whether the series converges absolutely, converges conditionally or diverges. Show all supporting work.

   a. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n\,(1+\sqrt{n})}$ b. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}\ln n}{n}$

7. (16 pts) Choose one. Find the convergence set for the power series. Show all supporting work.

   a. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}x^n}{n\ 2^n}$ b. $\sum_{n=1}^{\infty} \frac{5^n(x+1)^n}{n\,(n!)}$

8. (10 pts) Choose one. Determine how many terms of the series are necessary to estimate its sum to three-place accuracy. Using those terms, estimate the sum of the series.

   a. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}\log(n+80)}{8^n}$ b. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}3^n}{(2n)!}$